

Homework 5

Algebra Fluency: Distributing & Combining

Name: _____

Instructions

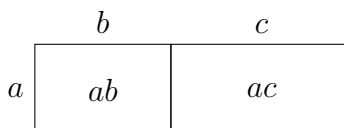
This homework will prepare you for the next stretch of calculus. Write your questions in the margin in a different color. You may use AI as a **teacher** (ask it to explain or give more examples), but do the problems yourself first.

Reading 1 — The Distributive Property

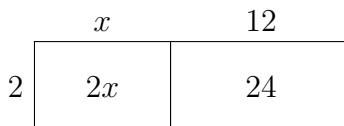
To **distribute** means to multiply a factor on the outside of a parenthesis by *every* term on the inside:

$$a(b + c) = ab + ac.$$

Why it works — think of area. A rectangle that is a tall and $(b + c)$ wide can be split into two smaller rectangles. Its total area is the sum of the two pieces:



The one rule: the outside factor hits **each** inside term. For example, $2(x + 12)$ is a 2-tall rectangle of width $(x + 12)$:



so $2(x + 12) = 2x + 24$. Another: $3(2x + 5) = 6x + 15$.

The tricky part: negatives. The sign on the outside travels onto *every* inside term.

$$-(x + 12) = -x - 12, \quad 2(x - 5) = 2x - 10, \quad -3(2x - 4) = -6x + 12.$$

In the last one, the -3 times the -4 gives $+12$ — two negatives make a positive. **Watch the signs: that is where most mistakes happen.**

Q1. Distribute each.

(a) $2(x + 3) =$ _____

(b) $5(x - 1) =$ _____

(c) $-(x + 4) =$ _____

(d) $-2(x - 6) =$ _____

(e) $3(2x + 5) =$ _____

(f) $-4(3x - 2) =$ _____

(g) $-(-x + 7) =$ _____

(h) $-5(2 - 3x) =$ _____

Reading 2 — Combining Like Terms

Like terms have the *same variable part* — the same letter raised to the same power. You can add or subtract like terms by combining their coefficients; unlike terms cannot be combined.

$$3x + 5x = 8x, \quad 7x - 2x = 5x, \quad 4x - 9x = -5x.$$

But $3x$ and $3x^2$ are **not** like terms (different powers), and $3x$ and 7 are **not** like terms (one has x , one does not). Watch the signs again: $4x - 9x = -5x$, not $5x$.

Putting it together: distribute, then combine.

$$2(x + 3) + 4x = 2x + 6 + 4x = 6x + 6.$$

Q2. Combine the like terms.

(a) $3x + 5x =$ _____

(b) $4x - 9x =$ _____

(c) $3x + 7 + 2x - 1 =$ _____

(d) $5x - 3 - 8x + 6 =$ _____

Q3. Distribute, then combine like terms.

(a) $2(x + 3) + 4x$

(b) $-2(x - 5) + 3x$

(c) $4(2x + 1) - 3(x - 2)$

Application — Areas and Catching Mistakes

Two-term products with the area model. To multiply $(x + 3)(x + 2)$, make a rectangle that is $(x + 3)$ wide and $(x + 2)$ tall, and split each side. Every piece of area is one product; add them all up.

	x	3
x	x^2	$3x$
2	$2x$	6

Adding the four pieces: $(x + 3)(x + 2) = x^2 + 3x + 2x + 6 = x^2 + 5x + 6$.

Q4. Minecraft floor. A room is $(x + 5)$ blocks wide and $(x + 2)$ blocks long. Fill in the area box, then write the total floor area as a single expanded expression.

	x	5
x		
2		

Total floor area = _____

Q5. Spot the error. Each line below has a mistake. Find it, say what went wrong in a few words, and write the correct answer.

(a) $-2(x - 5) = -2x - 10$

(b) $3(x + 4) = 3x + 4$

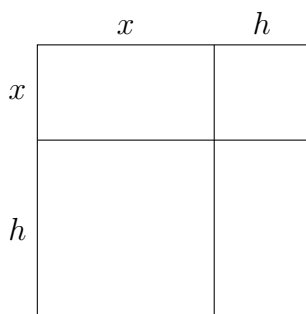
(c) $5x - 8x = 3x$

(d) $-(x - 7) = -x - 7$

★ Going Further (stretch)

This section is a stretch — give it a real try, and do not worry if you do not finish.

Squaring a sum. The area model also expands $(x + h)^2$ — a square that is $(x + h)$ on each side:



Q6. (a) Fill in the four pieces of the square above, then write $(x + h)^2$ as an expanded expression.

(b) Use it (or multiply it out) to expand $(x + h)^3 = (x + h)(x + h)^2$.

Q7. A tough simplification. Simplify this expression completely. Use your expansion of $(x + h)^2$ from Q6.

$$\frac{(x + h)^2 - x^2}{h}.$$